PROBLEM STATEMENT:

Customer support plays a crucial role in maintaining customer satisfaction and ensuring smooth communication between organizations and their users. In many organizations, customers raise issues, requests, or complaints through support systems in the form of **tickets**. These tickets must be assigned to the appropriate support agents who can analyze and resolve the problem efficiently. However, in many traditional support systems, the ticket assignment process is handled manually or through basic rules.

One of the major problems in existing support operations is the **manual assignment of tickets**. When a ticket is created, a supervisor or administrator may assign it to an available support agent. This process consumes time and requires continuous monitoring by the administrator.

Another major issue is the **uneven distribution of workload among support agents**. In many cases, some agents receive a large number of tickets while others receive very few. This imbalance occurs because the assignment process does not properly consider factors such as the current workload, availability, or expertise of the support agents. When some agents are overloaded with tickets, they may not be able to resolve issues quickly, which increases response time and reduces the overall efficiency of the support team.

The **lack of skill-based ticket assignment** is another important problem. Different support agents usually have different areas of expertise or technical knowledge. For example, some agents may specialize in network issues, while others may be more skilled in software-related problems or customer account management. In many traditional systems, tickets are assigned without considering these skill sets. As a result, an agent who does not have the appropriate expertise may receive the ticket, which may lead to delays, multiple transfers of the ticket between agents, and inefficient problem resolution.

In addition to these challenges, **priority handling of tickets** is also an important concern. Not all customer issues have the same level of urgency. Some tickets may involve critical system failures or urgent customer complaints that require immediate attention, while others may be routine service requests. Without a proper prioritization system, critical issues may not be addressed quickly. This can negatively affect service quality and may lead to dissatisfaction among customers.

Another problem in traditional support systems is the **lack of real-time monitoring and tracking of ticket assignments**. Administrators often find it difficult to track the status of all tickets, monitor the workload of agents, and ensure that issues are resolved within the expected time frame. Without proper tracking and reporting mechanisms, it becomes challenging to evaluate the performance of the support team and identify areas for improvement.

Furthermore, inefficient ticket assignment processes can lead to **poor customer experience**. When customers submit support requests, they expect quick responses and timely solutions. Delays in ticket assignment or incorrect routing of tickets can result in longer waiting times for customers. This may reduce trust in the organization's support services and negatively impact the organization's reputation.

As organizations continue to grow and customer demands increase, the volume of support tickets also increases significantly. Managing this growing number of tickets using manual processes becomes increasingly difficult. Therefore, organizations require a more **automated and intelligent system** that can handle ticket assignment efficiently and ensure that customer issues are addressed quickly.

To overcome these challenges, there is a need to develop a system that focuses on **streamlining the ticket assignment process for efficient support operations**. Such a system should automatically assign tickets to the most suitable support agents based on multiple factors .

In conclusion, the major problem addressed in this project is the **inefficient and manual ticket assignment process in support operations**, which leads to delays, workload imbalance, improper skill utilization, and reduced service quality.